Department of Physics, FNSPE, CTU in Prague

02YELMA - Electricity and Magnetism

Final Exam - Solutions 2

Q1: (a) Energy is conserved in the circuit, so the total voltage drop across the circuit elements must equal the applied voltage $V = V_R + V_L + V_C = 0$. The voltage drops are

$$V_R = RI, \quad V_L = L\frac{dI}{dt}, \quad V_C = \frac{q}{C}$$

The charge on the capacitor is related to the current by $I = dq/dt$. Thus,

$$RI + L\dot{I} + \frac{q}{C} = 0 \quad \xrightarrow{d/dt} \quad R\dot{I} + L\ddot{I} + \frac{1}{C}I = 0 \implies \boxed{\ddot{I} + \frac{R}{L}\dot{I} + \frac{1}{LC}I = 0}$$

(b) The governing equation describes a damped oscillation where the coefficient of the first derivative term determines the amount of damping: $2\beta = R/L$. To obtain the characteristic equation, we assume a solution of the form $I(t) = e^{\lambda t}$, which gives

$$\lambda^2 + \frac{R}{L}\lambda + \frac{1}{LC} = 0 \implies \lambda = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}.$$

For underdamped oscillations, the discriminant must be negative, which gives the condition on R, L, and C:

$$\boxed{R < 2\sqrt{L/C}}$$

For current to fall to half its initial value:

$$\frac{I_{\text{max}}e^{-\beta t}}{I_{\text{max}}} = \frac{1}{2} \implies e^{-\beta t} = \frac{1}{2} \implies \boxed{t = \frac{\ln 2}{\beta} = \frac{2L\ln 2}{R}}$$

(c) The total energy in the circuit is $U = U_L + U_C = \frac{1}{2}LI^2 + \frac{1}{2}CV^2$. If the capacitor is initially charged to V_0, then the initial energy is $U_0 = \frac{1}{2}CV_0^2$. When the current is at its maximum, the energy is stored entirely in the inductor, so $U = \frac{1}{2}LI_{\text{max}}^2$. At resonance, we know that only the resistor dissipates energy and current behaves as $I \sim I_0 \sin(\omega_0 t)$. Therefore, the average energy dissopated in the resistor over one cycle is

$$P_{\text{avg}} = \frac{1}{2}RI_0^2, \quad T = \frac{2\pi}{\omega_0} \implies \Delta E = P_{\text{avg}}T = \pi R I_0^2/\omega_0.$$

Therefore, the quality factor is

$$Q = 2\pi\frac{U}{\Delta E} = 2\pi\frac{\frac{1}{2}LI_0^2}{\pi RI_0^2/\omega_0} = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} \implies \boxed{Q = \frac{1}{R}\sqrt{\frac{L}{C}}}$$

Q2: (a) Because of cylindrical symmetry, the magnetic field can only have a angular component, $\vec{B}(\phi, t) = B(\phi, t)\,\hat{\phi}$. Consider an Amperian loop of radius $a < s < b$ coaxial with the shells:

$$B(\phi, t)\,(2\pi s) = \mu_0 I(t) \implies \boxed{\vec{B}(\phi, t) = \frac{\mu_0 I_0 \cos(\omega t)}{2\pi s}\,\hat{\phi}, \quad \text{for} \quad a < s < b.}$$

In region $s > b$, the net current enclosed by the Amperian loop is zero, so $\vec{B}(\phi, t) = 0$.
(b) The magnetic energy density is given by

$$u_B = \frac{B^2}{2\mu_0}.$$

The instantaneous magnetic energy stored per unit length is obtained by integrating over the vacuum region between the conductors:

$$\frac{U(t)}{L} = \int_a^b \frac{B^2(r,t)}{2\mu_0}\,(2\pi r\,dr) = \int_a^b \frac{1}{2\mu_0}\left(\frac{\mu_0 I(t)}{2\pi r}\right)^2 (2\pi r\,dr) = \frac{\mu_0 I^2(t)}{4\pi}\ln\left(\frac{b}{a}\right).$$

$$\Longrightarrow \boxed{\frac{U(t)}{L} = \frac{\mu_0 I_0^2}{4\pi}\cos^2(\omega t)\ln\left(\frac{b}{a}\right)}$$

The inductance per unit length is defined through

$$\frac{U}{L} = \frac{1}{2}\mathcal{L}I^2(t) \Longrightarrow \boxed{\mathcal{L} = \frac{\mu_0}{2\pi}\ln\left(\frac{b}{a}\right)}.$$

Q3: The solenoid is at rest in frame S' and produces the fields $\vec{E}' = 0,\ \vec{B}' = B_0\hat{z}'$. Frame S' moves relative to the laboratory frame S with velocity $\vec{v} = v\hat{x}$. Since the motion is along the x-direction, the field components parallel and perpendicular to $\vec{v}$ are $E'_\parallel = 0,\ E'_\perp = 0$ and because $\vec{B}'$ points along $\hat{z}$, $B'_\parallel = 0,\ B'_\perp = B_0\hat{z}$. Using

$$E_\parallel = E'_\parallel,\quad E_\perp = \gamma\left(E'_\perp - \vec{v}\times\vec{B}'\right) \qquad \Longrightarrow \qquad E_\parallel = 0,\quad \vec{E} = -\gamma(\vec{v}\times\vec{B}').$$

Compute the cross product: $\vec{v}\times\vec{B}' = (v\hat{x})\times(B_0\hat{z}) = vB_0(\hat{x}\times\hat{z}) = -vB_0\hat{y} \Longrightarrow \boxed{\vec{E} = \gamma v B_0\,\hat{y}}$.
Using

$$B_\parallel = B'_\parallel,\quad B_\perp = \gamma\left(B'_\perp + \frac{1}{c^2}\vec{v}\times\vec{E}'\right),\quad \vec{E}' = 0 \Longrightarrow B_\parallel = 0 \text{ and } \vec{B} = \gamma B_0\hat{z}.$$

$$\Longrightarrow \boxed{\vec{B} = \gamma B_0\,\hat{z}}.$$

Q4: The conducting rod of length L moves along horizontal rails and closes a circuit containing two resistors R (rod resistance) and R_0. The total resistance is $R_{\text{tot}} = R + R_0$. Let $x(t)$ denote the position of the rod, so the enclosed area is $A(t) = Lx(t)$. The magnetic flux through the loop is $\Phi_B(t) = B(t)Lx(t)$. Using Faraday's law, $\mathcal{E} = -\frac{d\Phi_B}{dt}$, we obtain

$$\mathcal{E} = -\frac{d}{dt}(BLx) = -L\left(x\frac{dB}{dt} + B\frac{dx}{dt}\right) \Longrightarrow \mathcal{E} = LB(t)\,(\alpha x - \dot{x}) \Longrightarrow \boxed{I(t) = \frac{LB_0 e^{-\alpha t}}{R+R_0}(\alpha x - \dot{x})}.$$

Direction follows Lenz's law:
- If $\alpha x > \dot{x}$, then $d\Phi_B/dt < 0$: the magnetic flux through the loop decreases, so the induced current must create magnetic field in the $+\hat{z}$ direction. Hence the current is **counterclockwise** (viewed from above).
- If $\alpha x < \dot{x}$, the flux increases and the current reverses: **clockwise**.
Initially ($v(0) = 0$):

$$I(0) = \frac{LB_0\alpha x_0}{R+R_0} > 0,$$

thus the initial current is counterclockwise.
(b) The rod experiences magnetic force $\vec{F} = I\,\vec{L}\times\vec{B}$. The rod carries current along $\hat{y}$, therefore $\vec{F} = ILB\,\hat{x}$. Substituting I,

$$F_x = \frac{L^2B^2}{R+R_0}(\alpha x - \dot{x}) \Longrightarrow m\ddot{x} = \frac{L^2B_0^2e^{-2\alpha t}}{R+R_0}(\alpha x - \dot{x}) \Longrightarrow \boxed{m\ddot{x} + \frac{L^2B_0^2e^{-2\alpha t}}{R+R_0}\dot{x} - \frac{\alpha L^2B_0^2e^{-2\alpha t}}{R+R_0}x = 0}$$

(c)

$$I(0) = \frac{LB_0\alpha x_0}{R+R_0} \Longrightarrow P_R(0) = I(0)^2R = \frac{L^2B_0^2\alpha^2x_0^2\,R}{(R+R_0)^2},\qquad P_{R_0}(0) = I(0)^2R_0 = \frac{L^2B_0^2\alpha^2x_0^2\,R_0}{(R+R_0)^2}$$

$$\Longrightarrow \boxed{P_{\text{tot}}(0) = P_R(0) + P_{R_0}(0) = \frac{L^2B_0^2\alpha^2x_0^2}{R+R_0}}$$